

ME 633: Design of Energy Systems

Course Code	ME 633
Course Name	Design of Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	B. Tech. /M. Tech./ MS/ PhD
Distribution	Compulsory for M. Tech in Mechanical Engineering with specialization in Energy Systems, and Elective for others
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**
Introduction to energy systems, Introduction to critical components for design in energy system, thermal stresses, material selections. [4 Lectures]
- **Module II**
Turbine blade design, overview of design criteria and certification guidelines, aerodynamic design, structural design, design and choice of sub-systems and components, design of blades for gas turbines, consideration of blade cooling in design of gas turbine blades, blades for hydro and wind turbines. [12 Lectures]
- **Module III**
Design of critical, supercritical and ultra-supercritical boilers, Stresses in pressure storage sections, Autofrettage, Thermal stresses, Design of various boiler components such as shell, heads, nozzles, flanges as per ASME & IS codes, Buckling. [8 Lectures]
- **Module IV**
Design of high pressure power plant piping systems, flow characteristics, material selection, thermal analysis of pipe and joints, thermal insulations. [4 Lectures]
- **Module V**
Design of heat transfer equipments, Boilers: classification, selection, Heat exchangers: classification, selection, flow friction and pressure drop analysis, basic thermal design, e-NTU, p-NTU, MTD methods. Shell and tube heat exchanger, construction and thermal features, thermal design procedure, Kern method, Bell Delaware method. [8 Lectures]
- **Module VI**
Thermal design of regenerators, classifications, design parameters. Design of compact heat exchangers, plate and fin, fin-tube and plate and frame heat exchangers, fouling and corrosion in heat exchangers. [6 Lectures]

Text and Reference Books

1. J. F. Harvey, **Theory and Design of Pressure Vessels**, CBS Publishers and Distributors, 1987.
2. S. Walas, **Chemical Process Equipment, Selection and Design**, Butterworths Sr. in Chemical Engineering.
3. L. Brownell, E. Young, **Process Equipment Design**, John Wiley and Sons.
4. D. Kern, **Process Heat Transfer**, Tata McGraw-Hill, 2000.
5. Fraas, **Heat Exchanger Design**, 2nd Edition, John Wiley & Sons, 1989.
6. J. Manwell, J. McGowan, A. Rogers, **Wind Energy Explained, Theory, Design and Application**, Wiley, 2012.
7. R. Kulwiec, **Materials Handling Handbook**, ASME, John Wiley and Sons.
8. W. Stoecker, **Design of thermal systems**, Tata McGraw-Hill Education, 3rd Edition.